

APPROVED METHOD:
PHOTOMETRIC TESTING OF FLOODLIGHTS
USING HIGH INTENSITY DISCHARGE OR
INCANDESCENT FILAMENT LAMPS
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-35-20(R2023)

**APPROVED METHOD:
PHOTOMETRIC TESTING OF FLOODLIGHTS
USING HIGH INTENSITY DISCHARGE
OR INCANDESCENT FILAMENT LAMPS
AN AMERICAN NATIONAL STANDARD**

Publication of this document
has been approved by the IES.
Suggestions for revisions should
be directed to the IES.

**Prepared by
The IES Testing Procedures Committee**



Copyright 2023 by the Illuminating Engineering Society.

Approved by the IES Standards Committee, December 19, 2022, as a Transaction of the Illuminating Engineering Society.

Approved April 28, 2023, as an American National Standard.

All rights reserved. No part of this publication may be reproduced in any form, in any electronic retrieval system or otherwise, without prior written permission of the IES.

Published by the Illuminating Engineering Society, 120 Wall Street, New York, New York 10005.

IES Standards are developed through committee consensus and produced by the IES Office in New York. Careful attention is given to style and accuracy. If any errors are noted in this document, they should be forwarded to the Director of Standards, at standards@ies.org or the above address for verification and correction. The IES welcomes and urges feedback and comments.

Printed in the United States of America.

ISBN# 978-0-87995-350-8

DISCLAIMER

IES publications are developed through the consensus standards development process approved by the American National Standards Institute. This process brings together volunteers representing varied viewpoints and interests to achieve consensus on lighting recommendations. While the IES administers the process and establishes policies and procedures to promote fairness in the development of consensus, it makes no guaranty or warranty as to the accuracy or completeness of any information published herein.

The IES disclaims liability for any injury to persons or property or other damages of any nature whatsoever, whether special, indirect, consequential or compensatory, directly or indirectly resulting from the publication, use of, or reliance on this document.

In issuing and making this document available, the IES is not undertaking to render professional or other services for or on behalf of any person or entity. Nor is the IES undertaking to perform any duty owed by any person or entity to someone else. Anyone using this document should rely on his or her own independent judgment or, as appropriate, seek the advice of a competent professional in determining the exercise of reasonable care in any given circumstances.

The IES has no power, nor does it undertake, to police or enforce compliance with the contents of this document. Nor does the IES list, certify, test or inspect products, designs, or installations for compliance with this document. Any certification or statement of compliance with the requirements of this document shall not be attributable to the IES and is solely the responsibility of the certifier or maker of the statement.

AMERICAN NATIONAL STANDARD

Approval of an American National Standard requires verification by ANSI that the requirements for due process, consensus, and other criteria have been met by the standards developer.

Consensus is established when, in the judgment of the ANSI Board of Standards Review, substantial agreement has been reached by directly and materially affected interests. Substantial agreement means much more than a simple majority, but not necessarily unanimity. Consensus requires that all views and objections be considered, and that a concerted effort be made toward their resolution.

The use of American National Standards is completely voluntary; their existence does not in any respect preclude anyone, whether that person has approved the standards or not, from manufacturing, marketing, purchasing, or using products, processes, or procedures not conforming to the standards.

The American National Standards Institute does not develop standards and will in no circumstances give an interpretation to any American National Standard. Moreover, no person shall have the right or authority to issue an interpretation of an American National Standard in the name of the American National Standards Institute. Requests for interpretations should be addressed to the secretariat or sponsor whose name appears on the title page of this standard.

CAUTION NOTICE: This American National Standard may be revised at any time. The procedures of the American National Standards Institute require that action be taken to reaffirm, revise, or withdraw this standard no later than five years from the date of approval. Purchasers of American National Standards may receive current information on all standards by calling or writing the American National Standards Institute.

Prepared by the IES Testing Procedures Committee

KC Fletcher, *Chair*

Andrew Jackson, *Vice Chair*

David N. Randolph, *Secretary*

Members

C. K. Andersen	M. L. Grather	J. E. Leland	S. Mitsuhashi
R. S. Bergman	J. N. Hulett	K. M. Liepmann	E. Radkov
B. Boudreaux	P.-C. Hung	J. Lockner	M. B. Sapcoe
E. Bretschneider	J. Jiao	S. Longo	J. S. Swiernik
F. Carpenter	M. Kotrebai	L. Loudin	J. C. Vollers
D. Ellis	B. Kuebler	J. P. Marella	

Advisory Members

L. M. Ayers	P. Elizondo	G. McKee	M. Ruffin
R. P. Bergin	A. A. Feldman	C. C. Miller	M. Schneider
C. A. Bloomfield	J. Grandusky	Y. Ohno	T. Schneider
P.-T. Chou	K. J. Hemmi	E. Page	A. W. Serres
T. E. Cowan	R. E. Hlgley	D. Park	P. Sharma
P. Cruz	G. John	E. S. Perkins	R. W. Siddon
L. Davis	J. Juhasz	M. Piscitelli	D. Spicer
F. De Caso	R. Kelley	D. P. Ramer	G. A. Steinberg
J. J. Demirjian	J. D. Kramer	T. W. Rasinsky	M. Stevenson
M. E. Duffy	K. C. Lerbs	D. Rogers	L. Swainston
V. Eberhard	P. McCarthy	M. Royer	A. Thorseth

CONTENTS

Foreword	1
1.0 Introduction and Scope	1
1.1 Introduction	1
1.2 Scope	1
2.0 Normative References	1
2.1 ANSI/IES LS-1-20	1
2.2 ANSI/IES LM-45-20	1
2.3 ANSI/IES LM-51-20	1
2.4 ANSI/IES LM-54-20	1
3.0 Nomenclature and Definitions	1
3.1 units	1
3.2 auxiliary equipment	1
4.0 Physical and Environmental Test Conditions	1
4.1 General	1
4.2 Temperature	2
4.3 Airflow	2
4.4 Stray Light	2
4.5 Vibration	2
5.0 Electrical Test Conditions	2
5.1 Power Supply Requirements	2
5.1.1 Primary Source Voltage	2
5.1.2 Power Supply Wave Shape	2
5.1.3 Power Supply Regulation	2
5.2 Reference Circuit Requirements	2
5.3 Measurement Instrument Requirements	2
5.3.1 Electrical Instruments	2

6.0	Testing Procedure Requirements	3
6.1	General Procedure	3
6.1.1	Goniophotometer	3
6.1.1.1	General	3
6.1.1.2	Coordinate Systems	3
6.1.1.3	Photometric System	3
6.1.1.4	Repeatability of Light Readings	4
6.2	Test Distance	4
6.3	Selection and Preparation of the Equipment for Test	4
6.3.1	Selection	4
6.3.2	Mounting	5
6.3.3	Lamp Orientation	5
6.3.3.1	Discharge Lamps	5
6.3.3.2	Incandescent Filament Lamps	6
6.3.4	Focusing	6
6.3.5	Cleaning	7
6.4	Test Lamps	7
6.4.1	General	7
6.4.2	Lamp Seasoning	7
6.4.3	Test Lamp Operation	7
6.4.3.1	General	7
6.4.3.2	Incandescent Filament Lamps	7
6.4.3.3	HID lamps	7
6.5	Test Methodologies	8
6.5.1	General	8
6.5.2	Bare Lamp Measurement	8
6.5.3	Lamp Photometric Calibration Process	8
6.5.3.1	Relative Photometry Method	8
6.5.3.1.1	Vertically Mounted Lamps	9
6.5.3.1.2	Horizontally Mounted Lamps	9
6.5.3.2	Absolute Photometry Method	9
6.5.3.2.1	Method Using Luminous Intensity Standard	9
6.5.3.2.2	Method Using Luminous Flux Standard	10
6.5.3.2.3	Method Using Calibrated Photometer	11
6.5.4	Luminaire Photometry	11
6.5.4.1	Mounting	11
6.5.4.2	Angles for Photometric Measurements	12
6.5.4.2.1	General	12
6.5.4.2.2	Standard Set of Lateral Data Angles	12
6.5.4.2.3	Standard Set of Vertical Data Angles	12
6.5.4.3	Photometric Data Processing	12
6.5.4.3.1	Stray Light Cancellation	12
6.5.4.3.2	Symmetry and Averaging of Data	12
6.5.4.3.3	Zonal Lumens and Total Lumens	12
6.5.4.3.4	Field Angle	12
6.5.4.3.5	Field Lumens	12

6.5.4.3.6	Beam Angle	13
6.5.4.3.7	Beam Lumens	13
6.5.4.3.8	Measuring and Recording Spill Light.....	13
6.5.4.3.9	Classification of Floodlight Beam Types.....	13
6.5.4.3.10	Isocandela Plot	13
7.0	Test Report	13
7.1	General	13
7.2	Test Report Contents	13
8.0	Additional Reading	14
Annex A	– Zonal Constants.....	14
Annex B	– Photometric Report and Test Results	17
Annex C	– Data File Formatted for Electronic Media.....	21
Informative References		24

Foreword

This approved method is a revision of and replacement for *LM-35-20/R23, IES Approved Method for Photometric Testing of Floodlights Using High Intensity Discharge or Incandescent Filament Lamps*. This version has been revised primarily to update the photometric, electrical, and environmental requirements in order to make them more consistent with current laboratory practice. The floodlight is aimed horizontally on either a C-type or B-type goniometer. The report (and *.IES photometric data file) continues to be presented in the traditional B-type format.

1.0 Introduction and Scope

1.1 Introduction

The objective of this Approved Method is to promote uniformity in measuring and reporting of the optical performance of floodlights and similar luminaires having a field angle equal to or greater than 10 degrees in any cross section.

1.2 Scope

This Approved Method applies to luminaires having a total field angle greater than or equal to 10 degrees. The method applies to luminaires that utilize a variety of light sources, including tungsten filament, tungsten-halogen, and high intensity discharge (HID) lamps. It does not apply to luminaires using low pressure sodium (LPS), fluorescent, or solid state lighting (SSL) sources. (For luminaires where the total field spread is less than 10 degrees, refer to ANSI/IES LM-11-20/R23.¹ Photometric testing of reflector-type lamps is described in ANSI/IES LM-20-20.²)

2.0 Normative References

2.1 ANSI/IES LS-1-20

Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: Illuminating Engineering Society; 2020.

2.2 ANSI/IES LM-45-20

Approved Method: Electrical and Photometric Measurement of General Service Incandescent Filament Lamps. New York: Illuminating Engineering Society; 2020.

2.3 ANSI/IES LM-51-20

Approved Method: Electrical and Photometric Measurement of High Intensity Discharge Lamps. New York: Illuminating Engineering Society; 2020.

2.4 ANSI/IES LM-54-20

Approved Method: IES Guide to Lamp Seasoning. New York: Illuminating Engineering Society; 2020.

3.0 Nomenclature and Definitions

The terms used in this document follow the definitions given in **Normative Reference 2.1**. Additional terms are defined in the subsections below.

3.1 units

The units of electrical measurement used in this Approved Method are the volt, the ampere, and the watt. The units of photometric measurement are the lumen and the candela. Color is specified in terms of the CIE recommended systems.

3.2 auxiliary equipment

The ballast and the igniter needed for the proper operation of discharge lamps.

4.0 Physical and Environmental Test Conditions

4.1 General

Changes in ambient temperature or air movement have little influence on the electrical and photometric characteristics of incandescent filament and HID lamps. In the case of HID lamps, the discharge tube is encapsulated inside an outer bulb, which provides a level of thermal and airflow isolation. Thus, the luminous

flux remains almost constant over a wide range of ambient temperatures.

It is good laboratory practice that the storage and testing of lamps be undertaken in a relatively clean environment. For lamps that operate at a high bulb temperature, contaminants can be etched into glass bulb surfaces during operation. Therefore, lamps should be cleaned before measurement.

4.2 Temperature

The ambient temperature of the photometric laboratory shall be maintained at $25\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$ ($77\text{ }^{\circ}\text{F} \pm 9\text{ }^{\circ}\text{F}$). The ambient temperature shall be measured at the same height and not more than 1.0 m (3.3 ft) from the luminaire. Temperature sensors shall be shielded from direct radiation and convection currents produced by the luminaire.

4.3 Airflow

Airflow is necessary to avoid thermal stratification and to control the ambient temperature. Good thermal control requires a high number of air exchanges per hour in the test area, but should be accomplished with low airflow rates.

4.4 Stray Light

Stray light should be suppressed in the test environment through the adequate use of low-reflectance surface finishes, shielding, and baffling. The walls, ceiling, and floor of the photometric test room should be painted flat black or covered with a low-reflectance matte material such as black felt, black velvet or black carpeting. Opaque low-reflectance baffles with limiting apertures interposed between the photodetector position and the luminaire position that shield the photodetector, except in the direction of the test source, should aid in suppressing the stray light reaching the photodetector. Stray light may be measured by running a complete or partial test with all direct light from the lamp or luminaire shielded from the photodetector.

4.5 Vibration

Lamps should not be subjected to excessive vibration or shock during measurement.

5.0 Electrical Test Conditions

5.1 Power Supply Requirements

5.1.1 Primary Source Voltage. In the case of HID lamps, ballasts should be operated at their nominal rated primary voltage and frequency. If the voltage rating is a range, the ballast should be operated at the intended application voltage, and the test voltage shall be recorded in the test report.

5.1.2 Power Supply Wave Shape. The AC power supply, during operation of the luminaire, shall have a sinusoidal voltage wave shape such that the value of the root-mean-square (RMS) summation of the harmonic components shall not exceed 3% of the fundamental. It is important to exercise caution when using constant-voltage transformer regulators. This type of system can produce harmonic distortion in excess of 3% when lightly loaded.

5.1.3 Power Supply Regulation. During testing, lamp wattage should be regulated to stabilize light output. Lamp wattage control is achieved via power supply regulation. The method used depends on the type of ballast and lamp used for testing. For incandescent lamps, lamp wattage is controlled via current regulation. For HID lamps running on electronic ballasts, the ballast electronics act as a wattage regulator, and only the supply voltage is regulated during testing. For HID lamps running on magnetic ballasts, wattage is regulated via supply voltage regulation. In all cases, to ensure stable luminous output from the lamp, the regulated parameter, either current or voltage, should be tightly regulated. (See **Section 6.4.3** for regulation details for each lamp type.)

5.2 Reference Circuit Requirements

There are no reference circuit requirements.

5.3 Measurement Instrument Requirements

5.3.1 Electrical Instruments. Electrical parameters (e.g., voltage, amperage, and wattage) shall be measured with instruments having traceability to the SI through a national metrology institute (NMI). The instrument

measurement tolerance of AC voltage and current shall be $\leq 0.2\%$. The instrument measurement tolerance of AC power shall be $\leq 0.5\%$. The instrument measurement tolerance of DC voltage and current shall be $\leq 0.1\%$. Instrument measurement ranges shall be selected so that the mid-to-maximum portion of the range is used for any specific measurement. Since gaseous discharge lamps may have seriously distorted wave shapes, AC instruments (for voltage, amperage, and/or wattage) shall measure true RMS values. Instruments with scales calibrated in RMS values but with deflections or readings based upon average or peak values shall not be used. If analog instrumentation is used, large-scale deflection for the intended measurement range should be used.

The electrical instruments should be chosen according to the guidelines and precautions presented in *ANSI/IES LM-28-20, Approved Method: IES Guide for the Selection, Care, and Use of Electrical Instruments in the Photometric Laboratory*.³ It should be noted that some HID electronic ballasts provide a high frequency lamp current. Care should be taken to ensure that possible conducted or radiated electromagnetic interference (EMI) from this type of ballast does not cause reading errors. High frequency wire runs should be as short as possible and, if necessary, shielded from EMI. It may be necessary to use precautions in order to prevent EMI from reaching voltage monitoring equipment or the photodetector readout device through the power circuits.

6.0 Testing Procedure Requirements

6.1 General Procedure

In general, photometric testing of floodlight luminaires should consist of:

- Selection of testing apparatus
- Selection and preparation of the luminaire to be tested
- Procedure for taking the measurements
- Computation of test results
- Reporting of data

6.1.1 Goniophotometer. An instrument for measuring the directional light distribution characteristics of

sources is called a goniophotometer. A goniophotometer system comprises a goniometer, a photometer and a subsystem for control and acquisition of data. The goniometer is the calibrated mechanical hardware that positions the source under test.

6.1.1.1 General. The goniometer provides a means of mounting the floodlight and rotating the floodlight through the required angular traverses. The goniometer shall be rigid enough to provide correct measurement of angles even when there is an appreciable off-balance loading. The goniometer should provide for accurate angular settings, and these should be reproducible to within 0.25 degrees.

6.1.1.2 Coordinate Systems. For many years, floodlights were tested using Type B goniophotometers, and the luminous intensity was presented in a Coordinate System B floodlight report. With this type of goniometer the lamp orientation changes with respect to gravity during the test. HID lamps are known to exhibit changes in lumen output as their operating position changes. Even incandescent lamps can experience changes in lumen output as a result of changing operating position. In addition, the changes in light center due to arc bowing in some HID lamps, and filament sag in some incandescent lamps, can cause the floodlight distribution to change as the luminaire changes orientation with respect to gravity. Because of these changes in light output during the photometric test, only Type-C goniometers shall be used for the testing of floodlights. Data shall be collected with the floodlight luminaire mounted on the Type-C goniometer and aimed horizontally. Collected data are subsequently converted to the equivalent Coordinate System B data.^{4,5}

6.1.1.3 Photometric System. The spectral response of the photometric system^{4,8,9} shall closely approximate the spectral luminous efficiency curve, $V(\lambda)$, for the CIE standard photometric observer. Silicon photodetectors are recommended because of their low noise and temperature sensitivity, excellent stability and linearity, fast response, and high sensitivity. A typical photometric detector is composed of a silicon photodiode, a $V(\lambda)$ -correction filter, and a precision aperture. Photodetectors shall be used within the limits of their rated output to avoid errors due to nonlinearity as the photocurrent approaches saturation. Use of a

calibrated photodetector, traceable to the SI through an NMI, is acceptable for the purpose of absolute and relative measurements.

The degree to which a photometric detector's spectral response approximates $V(\lambda)$ is described by the parameter f_1' . In general, the f_1' parameter compares the area under a detector's spectral responsivity curve to the area under the $V(\lambda)$ responsivity curve. A photodetector's f_1' value makes no guarantee about the differences in spectral responsivity in narrow spectral regions. Relative photometry on broadband sources, such as HID and incandescent lamps, is somewhat immune to the influences of the spectral content of the lamp and spectral filtering of the reflective and transmissive components in a luminaire.

The photometer head of the goniophotometer should have relative spectral responsivity matched to the $V(\lambda)$ function. The f_1' value of the spectral responsivity shall be less than 3%. For absolute measurements on spectrally structured sources (e.g., RGB or monochromatic) a systematic spectral correction factor should be considered. The correction factor is calculated from the spectral responsivity of the photometer head and the spectral distribution of the source to be measured. It is important to note that there are photometer heads readily available that have an f_1' value of spectral responsivity less than 1.5%.

Photocurrent in silicon photodetectors is typically linear over a dynamic range of approximately 10^6 . The absolute range for any given device is a function of the active area of the device and the electrical properties, such as the shunt resistance of the photodiode. The range of linear output for the device is typically defined as the range over which the photocurrent is linear within 1% and is significantly below the "saturation current" for the device. Care should be taken to ensure that the device is used within its linear range for the expected range of luminous intensity for the photometric measurements.

The calibration records for the photometric system should include the spectral response values of the detector from 380 to 780 nm, in increments no greater than 5 nm. The calibration documentation should also include the f_1' value.

6.1.1.4 Repeatability of Light Readings. One purpose of goniophotometry is to accurately measure the spatial distribution of light for a light source or luminaire. The goniophotometer should have good repeatability in light readings. There are many factors that contribute to uncertainty in light readings, and the goniophotometer should be evaluated to determine whether the system's measurement uncertainty is consistent with the measurement needs of the lamp or luminaire under test. Some factors influencing system uncertainty are:

- Positioning repeatability
- Accuracy of gain ratios in amplifier and scaling circuits
- Positioning accuracy
- Quantization error in analog-digital (A/D) conversion
- Linearity in the A/D conversion
- Spectral responsivity of the photodetector with respect to the source spectral properties

The uncertainty factors for any given goniophotometer depend on the design and construction of the instrument and the relative contributions of the factors appropriate for that system. As a general goal, light readings within a single test setup should have a repeatability of less than or equal to $\pm 2\%$.

6.2 Test Distance

The distance between the floodlight and the photodetector should be great enough that the inverse-square law applies. A distance of 7.5 to 10 m (25 to 33 ft) should be the minimum test distance for testing Types 4, 5, 6, or 7 floodlights (see **Table 6-2** in **Section 6.5.4.3.9**). The recommended test distance for Types 1, 2, or 3 may be longer to ensure meeting the inverse-square law requirement. (For more detail, refer to ANSI/IES LM-11-20.) In all cases, the test distance shall be recorded in the test report.

6.3 Selection and Preparation of the Equipment for Test

6.3.1 Selection. The floodlight being tested should represent a sample of the manufacturer's regular product. It should be checked to ensure that it is in good optical, mechanical, and electrical condition.

6.3.2 Mounting. The principal axis of a floodlight is defined as an imaginary line centered on the reflector opening and extending away from the floodlight perpendicular to the reflector opening (see **Figure 6-1**). The floodlight is aimed with its principal axis toward the horizon. The “top” of the floodlight will be oriented toward the goniometer’s zenith direction and the “bottom” will be oriented toward the goniometer’s nadir direction. The principal axis will be orientated toward the goniometer’s 0° lateral direction.

The floodlight should be positioned on the goniometer such that the center of the reflector opening coincides with the goniometer center.

6.3.3 Lamp Orientation. When a floodlight reflector and lamp are designed for a fixed relationship (as exists for a bi-post base lamp, a pre-focused base lamp, or a position-oriented mogul screw base for horizontal metal halide lamps), the normal positioning should be used for the test. When the relationship between reflector and lamp is not fixed (e.g., a standard mogul screw base lamp), the test arrangements described in **Sections 6.3.3.1** and **6.3.3.2** should be adopted and noted in

the test report (unless data are required at alternate lamp positions). These lamp orientations are chosen to permit averaging of the two sides of the beam values with minimum distortion of the beam shape and data. For configurations not described in **Section 6.3.3.1** or **6.3.3.2**, the lamp orientation should be observed and the position recorded in the test report.

6.3.3.1 Discharge Lamps. When a floodlight is designed around a discharge lamp, there are two variations in lamp orientation that are normally employed:

- For a lamp whose axis is perpendicular to the principal reflector axis, the arc tube support rods should lie in a plane parallel to the principal reflector axis. When only one support rod is present, it should be turned toward the reflector (see **Figure 6-2a**).
- For a lamp whose axis is parallel to the principal reflector axis, the arc tube support rod should be toward the top of the reflector. If two arc tube supports are used, they should be on the vertical center line (see **Figure 6-2b**).

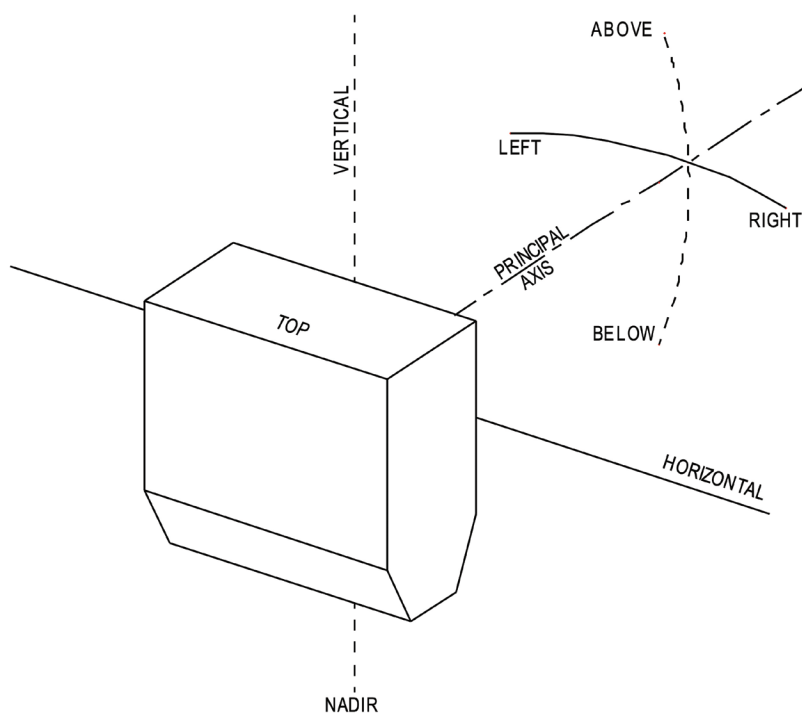


Figure 6-1. Defined directions around a floodlight. The principal axis is an imaginary line centered on (and perpendicular to) the reflector opening and extending away from the floodlight. (© Illuminating Engineering Society)

6.3.3.2 Incandescent Filament Lamps. When a floodlight is designed around an incandescent filament lamp, there are two variations in lamp orientation that are normally employed:

- For a lamp whose axis is perpendicular to the principal reflector axis, the open end of the filament should point away from the reflector (see **Figure 6-2c**).
- For a lamp whose axis is parallel to the principal reflector axis, the open end of the filament should point toward the top of the reflector (see **Figure 6-2d**).

When a filament lamp having a linear shaped filament (such as a CC8 filament) is used, it should be treated

in the same manner as discharge lamps (see **Section 6.3.3.1**).

6.3.4 Focusing. In fixed-focus floodlight units, the lamp light center should be placed at the designed focal point of the reflector. This often requires the careful measurement of the test lamp light center length and its location with respect to a designated reference point on the luminaire. The lamp position may need to be adjusted if the test lamp does not have the same light center length as a nominal lamp. In adjustable-focus units, the lamp should be adjusted in the floodlight to give the specific beam for which it is to be used. The lamp position used for the test shall be recorded in the test report.

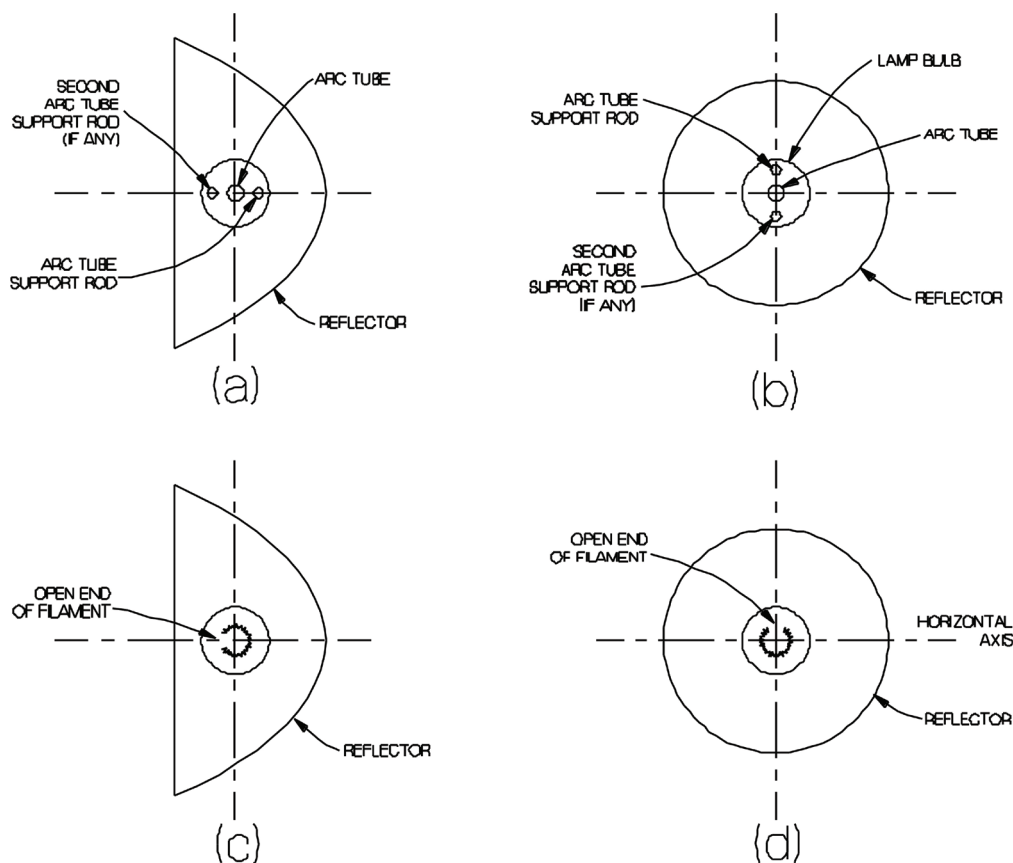


Figure 6-2. Lamp-reflector relationships for non-pre-focused lamps: (a) HID lamps with the lamp axis perpendicular to the principal floodlight axis (view looking along lamp axis); (b) HID lamps with the lamp axis parallel to the principal reflector axis (view looking in the front of the floodlight); (c) filament or tungsten-halogen lamps with the lamp axis perpendicular to the principal reflector axis (view looking along lamp axis); and (d) filament or tungsten-halogen lamps with the lamp axis parallel to the principal reflector axis (view looking in the front of the floodlight). (© Illuminating Engineering Society)

6.3.5 Cleaning. All glass, reflectors, and other optical parts should be thoroughly cleaned before any measurements are made unless the purpose of the test is to determine the effect of dirt on the floodlight. When cleaning a floodlight, it is desirable to return the floodlight to as near “like new” condition as possible, avoiding usage of corrosive or abrasive materials that might alter the finish of the lamp, reflector, refractor, or cover glass.

This can usually be accomplished by using a non-abrasive cleaner followed by much rinsing with distilled water. Cleaning methods and conditions should be noted in the test report.

6.4 Test Lamps

6.4.1 General. Lamps with a stable light output are needed for photometric test purposes and should remain so during a test or series of tests when constant line voltage is supplied. Test lamps should be selected for close conformance to the lamp manufacturer’s design dimensions and construction.

6.4.2 Lamp Seasoning. Test lamps shall be seasoned such that their characteristics remain constant during the test to be conducted. *ANSI/IES LM-54-20, Approved Method: Guide to Lamp Seasoning* (see **Section 2**) provides detailed information.

High intensity discharge (HID) lamps require special precautions when their operating orientation is changed. Long stabilization periods (up to several days) may be necessary to re-stabilize the lamps in the new orientation. Even designated “universal operating” HID lamps require prolonged stabilization in their actual operating position before being used as test lamps. In practice, separate lamps should be maintained for different orientations (base up, base down, or horizontal) and should be operated only in their assigned orientation. Lamp seasoning shall also be done in the same orientation as the intended orientation during testing.

6.4.3 Test Lamp Operation

6.4.3.1 General. During photometric testing, stable lamp performance is of primary importance. The

lamp is considered stabilized when the monitoring of light output over a period of 30 minutes produces fluctuations no greater than $\pm 1.0\%$ between any three consecutive readings taken approximately 15 minutes apart. Care shall be taken to be sure that the successive light readings are not increasing or decreasing monotonically. A false sense of stabilization can occur if the sampling period is short compared to the normal lamp stabilization time.

6.4.3.2 Incandescent Filament Lamps. A stabilization time of 10 minutes is usually sufficient for incandescent and tungsten-halogen lamps. Tungsten-halogen lamps shall be operated at a constant value of current near their nominal voltage. Other incandescent filament lamps should be operated at a constant current near 90% of rated input lamp voltage (while delivering approximately 75% of nominal luminous flux, this practice extends the life of test lamps). Current for incandescent lamps shall be regulated within $\pm 0.1\%$ (DC operation may be necessary to achieve this current regulation requirement). Operation at other voltages shall be specifically noted in the test report. When conducting absolute photometry, lamps should be operated under the conditions used to establish the nominal lumen rating.

For more information on electrical measurements, the reader should consult **Normative Reference 2.2**, *ANSI/ LM-45-20, Approved Method: Electrical and Photometric Measurement of General Service Incandescent Filament Lamps*.

6.4.3.3 HID lamps. A stabilization time may range from one to three hours for mercury, high pressure sodium, and metal halide lamps. HID lamps operating on magnetic ballasts shall be operated within $\pm 2.0\%$ of the rated power of the lamp and regulated to within $\pm 0.5\%$. HID lamps operating on electronic ballasts shall have the ballast input voltage set to within $\pm 0.5\%$ of the unit’s rated primary voltage and regulated to within $\pm 0.2\%$. If the rated primary voltage is a range, the test voltage shall be recorded in the test report. Metal halide-type HID lamps should not be moved after turn-off until the salts within the arc tube have cooled and solidified. For most applications, 30 minutes of cooling is sufficient.

For more information on electrical measurements, the reader should consult **Normative Reference 2.3**, *ANSI/ LM-51-20/R23, Approved Method: Electrical and Photometric Measurement of High Intensity Discharge Lamps*.

6.5 Test Methodologies

6.5.1 General. The lamp or luminaire shall be operated under conditions as typical of intended end use as possible, or suitable corrections made.

6.5.2 Bare Lamp Measurement. If a bare-lamp measurement is required (relative photometry), the following conditions shall be observed:

- a. The lamp(s) shall be mounted and operated in the same orientation as they would be in if they were in the luminaire. The light center of the test lamp (or the geometric center of the light centers of the test lamps) shall be mounted at the goniophotometer center.
- b. The same instruments (e.g., the photodetector and electrical measuring instruments) and test parameters (e.g., the test distance, electrical current for incandescent and tungsten-halogen lamps, and wattage for HID) shall be the same as used for the luminaire.
- c. Lamps and ballasts shall reach stable operating conditions before photometric readings are taken. Electronic monitoring should be used to determine when electrical and light readings become stable.
- d. Lamp lumen ratings should be taken from the current lamp schedule for the particular manufacturer's lamp used in the test.

6.5.3 Lamp Photometric Calibration Process. There are two basic methods to calibrate goniophotometric test data: relative photometry and absolute photometry. Each method has advantages and disadvantages.

6.5.3.1 Relative Photometry Method. In relative photometry, the goniophotometer is used to measure the test lamp independent of the luminaire to create a value typically referred to as the "Bare Lamp Sum." The Bare Lamp Sum is a number proportional to the lumen output of the lamp and is used to normalize the intensity data of

the luminaire. The normalized luminaire data can then be rescaled by the manufacturer's initial rated lumen output for the test lamp to create report data or IES photometric file data prorated for the lamp's rated lumen output. From a practical standpoint, relative photometry is desirable because a traceable photometric standard is not necessary.

The relative photometry method consists of the following steps:

1. Measure the test lamp (i.e., the "bare lamp"):
 - Measure the test lamp's luminous intensity distribution in the same orientation, under the same electrical operating conditions, and on the same goniophotometer as will be used for the luminaire test.
 - Use a measurement angle set appropriate for the lamp orientation. (See **Sections 6.5.3.1.1** and **6.5.3.1.2** for appropriate angle sets.)
2. Measure the luminaire:
 - Measure the luminaire's luminous intensity distribution using the same test lamp, under the same electrical operating conditions, and on the same goniophotometer as was used for the bare-lamp test.
 - Use a measurement angle set appropriate for the luminaire's expected photometric distribution type. (See **Section 6.4.1.3** for appropriate angle sets.)
3. Process the luminous intensity data:
 - Determine the ratio (R_G) of the bare-lamp test signal gain to the luminaire test signal gain.
 - Determine the manufacturer's initial lamp lumen rating for the test lamp.
 - Compute the test lamp's Bare Lamp Sum by summing the products of the measured intensity data and the zonal constants for the measurement angle set. (See **Annex A** for information on computing zonal constants.)
 - Scale the measured intensity data to create "corrected" luminous intensity values that are prorated to the test lamp's rated initial lumen output:

$$I_c = \frac{I_m \times LLR}{BLS} \times R_G$$

I_c = corrected luminous intensity value of luminaire at a given angle

I_m = measured luminous intensity value of luminaire at a given angle

BLS = Bare Lamp Sum

LLR = lamp lumen rating

R_G = ratio of measurement gains for the bare-lamp and luminaire tests

4. Compute the luminaire efficiency:

- Compute the "Luminaire Sum" by summing the products of the measured luminaire intensity data and the zonal constants for the luminaire's measurement angle set. (See **Annex A** for information on computing zonal constants.)
- Compute the efficiency:

$$E_L = \frac{L_S}{BLS} \times R_G$$

E_L = luminaire efficiency

L_S = Luminaire Sum

BLS = Bare Lamp Sum

R_G = ratio of measurement gains for the bare-lamp and luminaire tests

It is not necessary to measure the bare lamp before the luminaire.

6.5.3.1.1 Vertically Mounted Lamps. A vertically mounted (i.e., vertical base up or vertical base down) test lamp's luminous output shall be determined by measurements in at least 16 lateral half-planes spaced equally about the longitudinal lamp axis.

The vertical angle increments within the half-planes shall be sufficiently dense to represent the luminous intensity distribution such that each data point is a good approximation to the average luminous intensity within its zone. Vertical angles shall be 5, 15, 25, 35 ... 165, 175 for 10-degree zones. If necessary, finer angular increments may be used to properly represent the intensity distribution.

6.5.3.1.2 Horizontally Mounted Lamps. Horizontally mounted test lamps shall be calibrated in the end-use orientation. The lamp shall be positioned about its longitudinal axes in an orientation identical to that used for the luminaire. A horizontally mounted test lamp's luminous output shall be determined by measurements in at least 18

lateral half-planes spaced equally about the longitudinal lamp axis. Care should be taken to avoid measuring planes that may be shadowed by the goniophotometer support system or other structural items (e.g., 0 or 180 would typically contain a support structure for the lamp socket).

The vertical angle increments within the half-planes shall be sufficiently dense to represent the luminous intensity distribution such that each data point is a good approximation to the average luminous intensity within its zone. Vertical angles shall be 5, 15, 25, 35 ... 165, 175 for 10-degree zones. If necessary, finer angular increments may be used to properly represent the intensity distribution.

6.5.3.2 Absolute Photometry Method. In absolute photometry, the goniophotometer's responsivity in candelas per count (cd/count) is established using a mathematical model and system component data (e.g., mirror reflectivity, photodetector responsivity) or by measuring the system response to a traceable standard of luminous intensity or luminous flux. The calibrated goniophotometer can then be used to measure the properties of a light source or a luminaire in terms of absolute luminous intensity. Absolute photometry is advantageous or necessary when light source properties are strongly influenced by their use in the luminaire due to thermal effects such that the sources cannot be independently measured. Absolute photometry's drawback is that a traceable standard of luminous intensity is required to be maintained.

There are two methods for conducting absolute photometry of floodlight luminaires. In the first method, the goniophotometer detector system is calibrated using a standard of luminous intensity or a standard of luminous flux (a standard lamp). The second method is based on the use of a calibrated photometer.

6.5.3.2.1 Method Using Luminous Intensity Standard. A standard of luminous intensity is a light source that has been calibrated by a laboratory capable of establishing an absolute value of luminous intensity. A calibrated luminous intensity standard is mounted on the goniophotometer such that measurements will be made in the specified orientation and direction for which the luminous intensity standard is calibrated. The calibrated luminous intensity standard is powered in accordance with the instructions

provided by the calibration laboratory, and measurements are made using the goniophotometer's photodetector along the normal light path, inclusive of any system mirrors or other optical elements along the light path.

It is recommended that three standards be used. The ratio of the calibrated luminous intensity values to the measured values of the calibrated standards are averaged to establish a calibration factor for the goniophotometer.

An example calculation is provided using three luminous intensity standards:

<u>Luminous Intensity Standards</u>	<u>Photometer Readings</u>
Standard 1 = 1,500 cd	Standard 1: 3,782 counts
Standard 2 = 1,550 cd	Standard 2: 3,908 counts
Standard 3 = 1,532 cd	Standard 3: 3,862 counts
<u>Calibration Factor</u>	
Cal Factor 1 = $1,500/3,782 = 0.396615$	
Cal Factor 2 = $1,550/3,908 = 0.396622$	
Cal Factor 3 = $1,532/3,862 = 0.396685$	
Cal Factor Avg = $0.39664109 \text{ cd/count}$ (at measurement gain settings)	

This calibration factor is applied to the system gain setting at which the calibration measurements were made. For other system gain settings, the factor is scaled in proportion to the gain ratios. Goniophotometers require a wide dynamic range to handle the range of candela values created from bare lamps to beam-forming luminaires. Programmable amplifiers and high-resolution analog-to-digital converters normally provide this dynamic range. To establish an absolute system sensitivity that is independent of the gain settings, knowledge of the electrical signal processing is applied to compute the candelas per ampere of photocurrent.

6.5.3.2.2 Method Using Luminous Flux Standard. A *standard of luminous flux* is a light source that has been calibrated by a laboratory capable of establishing an absolute value of luminous flux. A calibrated luminous flux standard is mounted on the goniophotometer such that measurements will be made in the specified orientation for which the luminous flux standard is calibrated (i.e., typically vertical base up or vertical base down). The calibrated luminous flux standard is

powered in accordance with the instructions provided by the calibration laboratory, and measurements are made using the goniophotometer photodetector along the normal light path, inclusive of any system mirrors or other optical elements along the light path.

It is recommended that three standards be used. The ratio of the calibrated luminous flux values to the measured values of the calibrated standards are averaged to establish a calibration factor for the goniophotometer.

An example calculation is provided using luminous flux standards:

<u>Luminous Flux Standards</u>	<u>Photometer Readings</u>
Standard 1: 5,420 lm	Standard 1: 13,670 counts
Standard 2: 5,456 lm	Standard 2: 13,756 counts
Standard 3: 5,382 lm	Standard 3: 13,572 counts

In this example, the counts are representative of the "lumen sum" of the luminous flux standard. That is to say, the counts are representative of the sum of the products of the counts (proportional to the candela value) at each measurement angle times the zonal constant for the angular range of that measurement angle at the calibration gain setting.

<u>Calibration Factor</u>
Cal Factor = $5,420/13,670 = 0.3965$
Cal Factor = $5,456/13,756 = 0.3966$
Cal Factor = $5,382/13,572 = 0.3966$
Cal Factor = $0.3966 \text{ lm/(count}\cdot\text{sr)}$ (at the calibration gain setting)*

This Calibration Factor is applied to the system gain setting at which the calibration measurements were made. For other system gain settings, the factor is scaled in proportion to the gain ratios or calibrated directly at additional gain settings.

This System Calibration Factor (Sys Cal Factor) represents the system's sensitivity, inclusive of all optical factors, such as path length, mirror reflectivity, and photodetector responsivity, specific to the goniophotometer in use, and can be scaled as necessary to account for different signal processing settings, such as amplifier gains.

* cd = lm/sr

6.5.3.2.3 Method Using Calibrated Photometer. Absolute photometry using a calibrated photometer may be conducted by using the calibrated photometer as the detector system of the goniophotometer, or by transferring the calibration to the goniophotometer photodetector. The photometer's calibration should be valid over the full range of measurement values encountered during testing of the bare lamp and the luminaire; therefore, aspects such as detector linearity and spectral mismatch should be considered and preferably corrected.

The goniophotometer system calibration is determined by using the calibrated photodetector and the reflectance of any system mirrors or transmission of other optical elements along the light path under the intended conditions of use. For example, if the goniophotometer being calibrated uses a back surface mirror with source and detector aligned at 42 degrees from the mirror surface normal, then the mirror reflectance inclusive of the glass transmission losses should be known at that angle of incidence. This system calibration is used for the determination of total luminous flux of a bare lamp or luminaire. To determine the candela distribution of a source, the distance between the detector reference plane and the light source reference plane should be known. An example calculation is provided for both system calibrations (one for total luminous flux and one for luminous intensity).

Example 1: System Calibration in lx/count using a calibrated photometer:

A calibrated photometer traceable to the SI through an NMI corrected to the $V(\lambda)$ CIE photopic standard observer with $f_1' < 3\%$ has:

$$\text{Abs Cal} = 30.71 \text{ nA/lx} = 3.071 \times 10^{-8} \text{ A/lx} \\ (3.256 \times 10^7 \text{ lx/A}).$$

Gain Scaling Factor:

$$\text{Preamp transimpedance gain (G1)} = 1.0 \times 10^5 \text{ V/A}$$

$$\text{Post-amp gain (G2)} = 100 \text{ V/V}$$

$$\text{Gain factor} = G1 \times G2 = 1.0 \times 10^7 \text{ V/A} (1.0 \times 10^7 \text{ A/V})$$

Analog-to-digital converter parameters:

$$12\text{-bit A/D converter: } 4,096 \text{ counts full scale}$$

$$\text{Input signal range: } \pm 10 \text{ V full scale}$$

$$\text{A/D Factor: } 4.8828125 \text{ mV/count} (204.8 \text{ count/V})$$

Mirror reflectance (at angle of use):

$$\text{Reflectance} = 0.92 \text{ (for CIE illuminant A)}$$

It is important that the mirror reflectance be representative of the spectral and geometric conditions of use in the goniophotometric measurements.

Calibration Factor (for this gain setting):

$$\text{Cal Factor} = \text{Abs Cal} \times \text{Gain Factor} \times \text{A/D Factor} / \\ \text{Mirror Reflectance}$$

$$\text{Cal Factor} = 3.256 \times 10^7 \text{ lx/A} \times 1.0 \times 10^7 \text{ A/V} \times \\ 4.8828 \times 10^{-3} \text{ V/count} / 0.92$$

$$\text{Cal Factor} = 1.728 \times 10^{-2} \text{ lx/count}$$

Example 2: System Calibration in cd/count using a calibrated photometer:

Candela Calibration Factor (for this gain setting):

$$\text{Cd Cal Factor} = \text{Cal Factor (see previous example)} \times \\ \text{Distance}^2 (L1+L2)$$

$$\text{Cd Cal Factor} = 1.728 \times 10^{-2} \text{ lx/count} \times (8.14 \text{ m}^2)$$

$$\text{Cd Cal Factor} = 1.145 \text{ cd/count}$$

The test lamp is operated under the same electrical operating conditions as will be used in testing the luminaire. The selected current, wattage, or voltage shall be held constant throughout the test.

The luminous intensity distribution of the luminaire, with the test lamp installed in the same orientation and operated under the same electrical conditions as were used when the luminous flux of the test lamp was established, shall then be measured with the calibrated goniophotometer.

For absolute photometry, the calibration of the standard lamp or the standard photometer shall be traceable to the SI through a national metrology institute (NMI). In addition, the laboratory should determine the measurement uncertainty when transferring standard values.

6.5.4 Luminaire Photometry. In addition to the information provided in this section,⁵ other information is available.⁷

6.5.4.1 Mounting. The luminaire should be mounted on a goniophotometer that is designed for minimum obstruction of light. Care should be exercised to prevent

light from reflecting off of the mounting apparatus. If this light is redirected directly into the photodetector or back onto the luminaire, it may artificially increase luminaire intensities.

6.5.4.2 Angles for Photometric Measurements

6.5.4.2.1. General. A standard set of data angles is established to be used for all beam spreads. Near the floodlight principal axis the data points are close together and spread further apart as they approach 90° from the principal axis. For narrow beam spreads, there are adequate points to define the beam. For wider beams, the beam edges have adequate data points. By using this set of data angles, the historic practice of trial measurement of beam width before picking a set of angles for data recording is avoided.

6.5.4.2.2 Standard Set of Lateral Data Angles. As a minimum, data angles shall be: 0, 1, 3, 5, 7, 9, 11, 13, 15, 17, 19.5, 22.5, 25.5, 29, 33, 37.5, 42.5, 47.5, 55, 65, 75, 85, and 90 degrees on either side of the principal axis. This is a total of 45 angles. Zero degrees is at the principal axis, and 90 degrees is perpendicular to the principal axis. The data angles that are used for lumen calculations are shown in **Table 6-1**.

Table 6-1. Data Angles Used for Lumen Calculations

Angle from Principal Axis (degrees)						
1	3	5	7	9	11	13
15	17	19.5	22.5	25.5	29	33
37.5	42.5	47.5	55	65	75	85

Table notes:

Angles 1 thru 17 are mid-zone angles for 2-degree zones
 Angles 19.5 thru 25.5 are mid-zone angles for 3-degree zones
 Angles 29 thru 33 are mid-zone angles for 4-degree zones
 Angles 37.5 thru 47.5 are mid-zone angles for 5-degree zones
 Angles 55 thru 85 are mid-zone angles for a 10-degree zone

6.5.4.2.3 Standard Set of Vertical Data Angles. Most modern Type-C goniophotometers can acquire data in small vertical increments without causing a significant increase in test time. If this is the case, it may be desirable to acquire data in 0.5-degree increments in the vertical plane to ensure that the full set of required angles is acquired. In any case, the minimum set of vertical data angles in Type C coordinates is: 0, 5, 15, 25,

35, 42.5, 47.5, 52.5, 57, 61, 64.5, 67.5, 70.5, 73, 75, 77, 79, 81, 83, 85, 87, 89, 90, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109.5, 112.5, 115.5, 119, 123, 127.5, 132.5, 137.5, 145, 155, 165, 175, and 180 degrees. This is a total of 45 vertical angles from nadir to zenith.

6.5.4.3 Photometric Data Processing

6.5.4.3.1 Stray Light Cancellation. Stray light may be measured by running a complete or partial test with all direct light from the lamp or luminaire shielded from the photodetector. The choice of partial or complete measurement of the stray light depends on the spatial properties of the luminaire's photometric pattern and the expected stray light contribution at each measured vertical and horizontal angle pair. The measured stray light, if any, shall then be subtracted from the bare lamp or luminaire test data, at each vertical angle in each plane measured.

6.5.4.3.2 Symmetry and Averaging of Data. Many floodlight luminaires exhibit bilateral symmetry about the zero-degree lateral plane. In this case, a full set of data shall be taken, but averages may be made of the values taken at corresponding angles on either side of the plane of symmetry.

6.5.4.3.3 Zonal Lumens and Total Lumens. The luminous flux (lumens) in each test zone is calculated by using the measured luminous intensity and appropriate zonal constant (see **Annex A**) for the zone. Total luminous flux (in lumens) is calculated by summing all of the zonal lumen values.

6.5.4.3.4 Field Angle. The field angle is the difference in degrees between the 10% points of maximum luminous intensity on either side of the maximum intensity point. The field angle is determined for both the horizontal and vertical axes of the floodlight. The field angles for the two directions are usually different. When the field angle does not occur on the axis, the field angle and its location should be noted.

6.5.4.3.5 Field Lumens. Field lumens are calculated by summing the number of lumens in those test zones that have a central luminous intensity greater than or equal to 10% of the maximum luminous intensity.

6.5.4.3.6 Beam Angle. The beam angle is the number of degrees between the 50% points of maximum luminous intensity on either side of the maximum intensity point. The beam angle is determined for both the horizontal and vertical axes of the floodlight. The beam angles for the two directions are usually different. When the beam angle does not occur on the axis, the beam angle and its location should be noted.

6.5.4.3.7 Beam Lumens. Beam lumens are calculated by summing the number of lumens in those test zones that have a central luminous intensity greater than or equal to 50% of the maximum luminous intensity.

6.5.4.3.8 Measuring and Recording Spill Light. Spill light is defined as that light emitted by the floodlight that is outside the floodlight distribution as defined by the field angle classification. Using the standard set of data angles ensures that light beyond the field angle is recorded out to 90° from the principal reflector axis.

6.5.4.3.9 Classification of Floodlight Beam Types. Floodlight beam types are typically classified by an IES type in both the horizontal and vertical axes. This classification is based on the field angle of the luminaire in both the horizontal and vertical directions. **Table 6-2** shows the classifications and their corresponding field angles. The designation is shown as H x V; for example, a flood with a horizontal (H) field angle of 45 degrees and a vertical (V) field angle of 61 degrees would have a 3 x 4 beam type.

Table 6-2. Floodlight Designations

IES Type	Field Angle (degrees)
1	10 – 18
2	18 – 29
3	29 – 46
4	46 – 70
5	70 – 100
6	100 – 130
7	130 – 180

6.5.4.3.10 Isocandela Plot. A plot of isolines of constant luminous intensity (either in candelas or as a percentage of the maximum candelas) shall be plotted on a rectangular plot with vertical angles as the ordinate axis and horizontal angles as the abscissa.

7.0 Test Report

7.1 General

The test report shall describe all significant data concerning the luminaire tested, and the data should be listed so that the maximum amount of information can be interpreted. **Section 7.2** identifies data items to be included in the test report. The sample report in **Annex B** illustrates typical data and information content presented in a floodlight test report.

7.2 Test Report Contents

Test reports shall include all data and descriptive information necessary to adequately describe the luminaire under test and the test conditions. The goal of providing this information is that users of the report will have sufficient information to verify that the contents of the report are relevant to the planned use of both the report and the associated "IES file" data. While the exact information recorded is specific to the devices being tested, and the format is somewhat specific to the testing agency, typical data included in the test report are:

- General
 - Name of testing agency
 - Test number
 - Date of test
- Floodlight description
 - Manufacturer's name
 - Floodlight type, catalog number, description to identify floodlight
 - Sketch of floodlight showing size, dimensions, and location on the goniometer
 - Reflector shape, material, and dimensions
- Lamp description
 - ANSI type, order abbreviation
 - Rating in watts and lumens
 - Bulb shape, size, finish, and base type
 - Filament or arc tube construction and light center length
 - Specification of support or lead wire positions
 - Light center position relative to the reflector during test

- Distribution characteristics
 - IES type, both horizontal and vertical
 - Maximum luminous intensity value and its location
 - Field angle in both horizontal and vertical directions
 - Beam angle in both horizontal and vertical directions
 - Field and beam flux (lumens) and field efficiency
 - Spill light lumens
 - Total flux and total efficiency
- Distribution data
 - Iso-candela curves (showing equal luminous intensities; averaged for right and left sides when symmetrical)
 - Horizontal and vertical intensity distribution curves passing through 0° H, 0° V
- Other data
 - A statement about the test distance; for example: "This test was performed in accordance with ANSI/IES LM-35-20 with a test distance of 8 meters (26 feet)"
 - Tabulations of candela values and zonal lumen values (optional)

A copy of the photometry and related data may be created in an electronic file. The file shall be formatted per *ANSI/IES LM-63-19, IES Standard File Format for Electronic Transfer of Photometric Data*.⁸ (See **Annex C** for an example of a formatted file.)

8.0 Additional Reading

Barclay DN. Photometric characteristics of metal halide lamps. *J Illum Engineering Soc.* 1977;7(3):197-200.

Illuminating Engineering Society. ANSI/IES LM-52-03(R2014), Photometric Measurements of Roadway Sign Installations. New York: IES; 2014.

Illuminating Engineering Society. ANSI/IES LS-4-20, Lighting Science: Measurement of Light – The Science of Photometry. New York: IES; 2020.

International Standards Organization (ISO) and International Commission on Illumination (CIE). ISO/CIE 19476:2014(E), Methods of Characterizing Illuminance Meters and Luminance Meters: Performance Characteristics and Specifications. Vienna: CIE; 2014.

Ohno Y, Cromer CL, Hardis JE, Eppeldauer G. The detector-based candela scale and related photometric calibration procedures at NIST. *J Illum Engineering Soc.* 1994;23(1):89-98.

Walsh JWT. Photometry, 3rd ed. London: Constable and Co., Ltd; 1958.

Annex A – Zonal Constants

The product of the mid-zone candela and the zonal constant (solid-angle volume in steradians) gives the zonal lumens. The summation of the zonal lumens multiplied by 100 and divided by the nominal lamp lumens gives the percent efficiency of the floodlight. The formula for determining the zonal constant (K) is:

$$K = [\phi \pi (\sin \theta_1 - \sin \theta_2)] / 180$$

where:

ϕ = interval, in degrees, around the polar axis

θ_1 and θ_2 = limits of the angle, in degrees, from the polar axis

The formula is convenient to use in computer programming.

Tables of constants for standard data angles are included in this Annex. **Table A-1** is for Type-B coordinates, and **Table A-2** is for Type-C coordinates.

**Approved Method: Photometric Testing of Floodlights Using High Intensity
Discharge or Incandescent Filament Lamps**

Table A-1. Zonal Lumen Constants for Converting Candelas to Lumens for Type B Coordinates

85	0.0060911	0.0060837	0.0060689	0.0060466	0.0060170	0.0059801	0.0059359	0.0058845	0.0058258	0.0086134	0.0084419
75	0.0060911	0.0060837	0.0060689	0.0060466	0.0060170	0.0059801	0.0059359	0.0058845	0.0058258	0.0086134	0.0084419
65	0.0060911	0.0060837	0.0060689	0.0060466	0.0060170	0.0059801	0.0059359	0.0058845	0.0058258	0.0086134	0.0084419
55	0.0060911	0.0060837	0.0060689	0.0060466	0.0060170	0.0059801	0.0059359	0.0058845	0.0058258	0.0086134	0.0084419
47.5	0.0030456	0.0030418	0.0030344	0.0030233	0.0030085	0.0029901	0.0029680	0.0029422	0.0029129	0.0043067	0.0042210
42.5	0.0030456	0.0030418	0.0030344	0.0030233	0.0030085	0.0029901	0.0029680	0.0029422	0.0029129	0.0043067	0.0042210
37.5	0.0030456	0.0030418	0.0030344	0.0030233	0.0030085	0.0029901	0.0029680	0.0029422	0.0029129	0.0043067	0.0042210
33	0.0024364	0.0024335	0.0024275	0.0024187	0.0024068	0.0023920	0.0023744	0.0023538	0.0023303	0.0034453	0.0033768
29	0.0024364	0.0024335	0.0024275	0.0024187	0.0024068	0.0023920	0.0023744	0.0023538	0.0023303	0.0034453	0.0033768
25.5	0.0018273	0.0018251	0.0018207	0.0018140	0.0018051	0.0017940	0.0017808	0.0017653	0.0017478	0.0025840	0.0025326
22.5	0.0018273	0.0018251	0.0018207	0.0018140	0.0018051	0.0017940	0.0017808	0.0017653	0.0017478	0.0025840	0.0025326
19.5	0.0018273	0.0018251	0.0018207	0.0018140	0.0018051	0.0017940	0.0017808	0.0017653	0.0017478	0.0025840	0.0025326
17	0.0012182	0.0012167	0.0012138	0.0012093	0.0012034	0.0011960	0.0011872	0.0011769	0.0011652	0.0017227	0.0016884
15	0.0012182	0.0012167	0.0012138	0.0012093	0.0012034	0.0011960	0.0011872	0.0011769	0.0011652	0.0017227	0.0016884
13	0.0012182	0.0012167	0.0012138	0.0012093	0.0012034	0.0011960	0.0011872	0.0011769	0.0011652	0.0017227	0.0016884
11	0.0012182	0.0012167	0.0012138	0.0012093	0.0012034	0.0011960	0.0011872	0.0011769	0.0011652	0.0017227	0.0016884
9	0.0012182	0.0012167	0.0012138	0.0012093	0.0012034	0.0011960	0.0011872	0.0011769	0.0011652	0.0017227	0.0016884
7	0.0012182	0.0012167	0.0012138	0.0012093	0.0012034	0.0011960	0.0011872	0.0011769	0.0011652	0.0017227	0.0016884
5	0.0012182	0.0012167	0.0012138	0.0012093	0.0012034	0.0011960	0.0011872	0.0011769	0.0011652	0.0017227	0.0016884
3	0.0012182	0.0012167	0.0012138	0.0012093	0.0012034	0.0011960	0.0011872	0.0011769	0.0011652	0.0017227	0.0016884
1	0.0012182	0.0012167	0.0012138	0.0012093	0.0012034	0.0011960	0.0011872	0.0011769	0.0011652	0.0017227	0.0016884
Type B Vertical Angles*	1	3	5	7	9	11	13	15	17	19.5	22.5
	Horizontal Mid-zone Angles from the Principal Axis										

85	0.0082474	0.0106548	0.0102169	0.0120796	0.0112258	0.0102866	0.0174500	0.0128574	0.0078741	0.0026515
75	0.0082474	0.0106548	0.0102169	0.0120796	0.0112258	0.0102866	0.0174500	0.0128574	0.0078741	0.0026515
65	0.0082474	0.0106548	0.0102169	0.0120796	0.0112258	0.0102866	0.0174500	0.0128574	0.0078741	0.0026515
55	0.0082474	0.0106548	0.0102169	0.0120796	0.0112258	0.0102866	0.0174500	0.0128574	0.0078741	0.0026515
47.5	0.0041237	0.0053274	0.0051084	0.0060398	0.0056129	0.0051433	0.0087250	0.0064287	0.0039370	0.0013258
42.5	0.0041237	0.0053274	0.0051084	0.0060398	0.0056129	0.0051433	0.0087250	0.0064287	0.0039370	0.0013258
37.5	0.0041237	0.0053274	0.0051084	0.0060398	0.0056129	0.0051433	0.0087250	0.0064287	0.0039370	0.0013258
33	0.0032989	0.0042619	0.0040867	0.0048319	0.0044903	0.0041146	0.0069800	0.0051429	0.0031496	0.0010606
29	0.0032989	0.0042619	0.0040867	0.0048319	0.0044903	0.0041146	0.0069800	0.0051429	0.0031496	0.0010606
25.5	0.0024742	0.0031964	0.0030651	0.0036239	0.0033677	0.0030860	0.0052350	0.0038572	0.0023622	0.0007955
22.5	0.0024742	0.0031964	0.0030651	0.0036239	0.0033677	0.0030860	0.0052350	0.0038572	0.0023622	0.0007955
19.5	0.0024742	0.0031964	0.0030651	0.0036239	0.0033677	0.0030860	0.0052350	0.0038572	0.0023622	0.0007955
17	0.0016495	0.0021310	0.0020434	0.0024159	0.0022452	0.0020573	0.0034900	0.0025715	0.0015748	0.0005303
15	0.0016495	0.0021310	0.0020434	0.0024159	0.0022452	0.0020573	0.0034900	0.0025715	0.0015748	0.0005303
13	0.0016495	0.0021310	0.0020434	0.0024159	0.0022452	0.0020573	0.0034900	0.0025715	0.0015748	0.0005303
11	0.0016495	0.0021310	0.0020434	0.0024159	0.0022452	0.0020573	0.0034900	0.0025715	0.0015748	0.0005303
9	0.0016495	0.0021310	0.0020434	0.0024159	0.0022452	0.0020573	0.0034900	0.0025715	0.0015748	0.0005303
7	0.0016495	0.0021310	0.0020434	0.0024159	0.0022452	0.0020573	0.0034900	0.0025715	0.0015748	0.0005303
5	0.0016495	0.0021310	0.0020434	0.0024159	0.0022452	0.0020573	0.0034900	0.0025715	0.0015748	0.0005303
3	0.0016495	0.0021310	0.0020434	0.0024159	0.0022452	0.0020573	0.0034900	0.0025715	0.0015748	0.0005303
1	0.0016495	0.0021310	0.0020434	0.0024159	0.0022452	0.0020573	0.0034900	0.0025715	0.0015748	0.0005303
Type B Vertical Angles*	25.5	29	33	37.5	42.5	47.5	55	65	75	85
	Horizontal Mid-zone Angles from the Principal Axis									

*Vertical mid-zone angles from the principal axis

Table A-2. Zonal Lumen Constants for Converting Candelas to Lumens for Type C Coordinates

175	0.0005303	0.0005303	0.0005303	0.0005303	0.0005303	0.0005303	0.0005303	0.0005303	0.0005303	0.0007955	0.0007955
165	0.0015748	0.0015748	0.0015748	0.0015748	0.0015748	0.0015748	0.0015748	0.0015748	0.0015748	0.0023622	0.0023622
155	0.0025715	0.0025715	0.0025715	0.0025715	0.0025715	0.0025715	0.0025715	0.0025715	0.0025715	0.0038572	0.0038572
145	0.0034900	0.0034900	0.0034900	0.0034900	0.0034900	0.0034900	0.0034900	0.0034900	0.0034900	0.0052350	0.0052350
137.5	0.0020573	0.0020573	0.0020573	0.0020573	0.0020573	0.0020573	0.0020573	0.0020573	0.0020573	0.0030860	0.0030860
132.5	0.0022452	0.0022452	0.0022452	0.0022452	0.0022452	0.0022452	0.0022452	0.0022452	0.0022452	0.0033677	0.0033677
127.5	0.0024159	0.0024159	0.0024159	0.0024159	0.0024159	0.0024159	0.0024159	0.0024159	0.0024159	0.0036239	0.0036239
123	0.0020434	0.0020434	0.0020434	0.0020434	0.0020434	0.0020434	0.0020434	0.0020434	0.0020434	0.0030651	0.0030651
119	0.0021310	0.0021310	0.0021310	0.0021310	0.0021310	0.0021310	0.0021310	0.0021310	0.0021310	0.0031964	0.0031964
115.5	0.0016495	0.0016495	0.0016495	0.0016495	0.0016495	0.0016495	0.0016495	0.0016495	0.0016495	0.0024742	0.0024742
112.5	0.0016884	0.0016884	0.0016884	0.0016884	0.0016884	0.0016884	0.0016884	0.0016884	0.0016884	0.0025326	0.0025326
109.5	0.0017227	0.0017227	0.0017227	0.0017227	0.0017227	0.0017227	0.0017227	0.0017227	0.0017227	0.0025840	0.0025840
107	0.0011652	0.0011652	0.0011652	0.0011652	0.0011652	0.0011652	0.0011652	0.0011652	0.0011652	0.0017478	0.0017478
105	0.0011769	0.0011769	0.0011769	0.0011769	0.0011769	0.0011769	0.0011769	0.0011769	0.0011769	0.0017653	0.0017653
103	0.0011872	0.0011872	0.0011872	0.0011872	0.0011872	0.0011872	0.0011872	0.0011872	0.0011872	0.0017808	0.0017808
101	0.0011960	0.0011960	0.0011960	0.0011960	0.0011960	0.0011960	0.0011960	0.0011960	0.0011960	0.0017940	0.0017940
99	0.0012034	0.0012034	0.0012034	0.0012034	0.0012034	0.0012034	0.0012034	0.0012034	0.0012034	0.0018051	0.0018051
97	0.0012093	0.0012093	0.0012093	0.0012093	0.0012093	0.0012093	0.0012093	0.0012093	0.0012093	0.0018140	0.0018140
95	0.0012138	0.0012138	0.0012138	0.0012138	0.0012138	0.0012138	0.0012138	0.0012138	0.0012138	0.0018207	0.0018207
93	0.0012167	0.0012167	0.0012167	0.0012167	0.0012167	0.0012167	0.0012167	0.0012167	0.0012167	0.0018251	0.0018251
91	0.0012182	0.0012182	0.0012182	0.0012182	0.0012182	0.0012182	0.0012182	0.0012182	0.0012182	0.0018273	0.0018273
Type C	1	3	5	7	9	11	13	15	17	19.5	22.5

175	0.0007955	0.0010606	0.0010606	0.0013258	0.0013258	0.0013258	0.0026515	0.0026515	0.0026515	0.0026515	5
165	0.0023622	0.0031496	0.0031496	0.0039370	0.0039370	0.0039370	0.0078741	0.0078741	0.0078741	0.0078741	15
155	0.0038572	0.0051429	0.0051429	0.0064287	0.0064287	0.0064287	0.0128574	0.0128574	0.0128574	0.0128574	25
145	0.0052350	0.0069800	0.0069800	0.0087250	0.0087250	0.0087250	0.0174500	0.0174500	0.0174500	0.0174500	35
137.5	0.0030860	0.0041146	0.0041146	0.0051433	0.0051433	0.0051433	0.0102866	0.0102866	0.0102866	0.0102866	42.5
132.5	0.0033677	0.0044903	0.0044903	0.0056129	0.0056129	0.0056129	0.0112258	0.0112258	0.0112258	0.0112258	47.5
127.5	0.0036239	0.0048319	0.0048319	0.0060398	0.0060398	0.0060398	0.0120796	0.0120796	0.0120796	0.0120796	52.5
123	0.0030651	0.0040867	0.0040867	0.0051084	0.0051084	0.0051084	0.0102169	0.0102169	0.0102169	0.0102169	57
119	0.0031964	0.0042619	0.0042619	0.0053274	0.0053274	0.0053274	0.0106548	0.0106548	0.0106548	0.0106548	61
115.5	0.0024742	0.0032989	0.0032989	0.0041237	0.0041237	0.0041237	0.0082474	0.0082474	0.0082474	0.0082474	64.5
112.5	0.0025326	0.0033768	0.0033768	0.0042210	0.0042210	0.0042210	0.0084419	0.0084419	0.0084419	0.0084419	67.5
109.5	0.0025840	0.0034453	0.0034453	0.0043067	0.0043067	0.0043067	0.0086134	0.0086134	0.0086134	0.0086134	70.5
107	0.0017478	0.0023303	0.0023303	0.0029129	0.0029129	0.0029129	0.0058258	0.0058258	0.0058258	0.0058258	73
105	0.0017653	0.0023538	0.0023538	0.0029422	0.0029422	0.0029422	0.0058845	0.0058845	0.0058845	0.0058845	75
103	0.0017808	0.0023744	0.0023744	0.0029680	0.0029680	0.0029680	0.0059359	0.0059359	0.0059359	0.0059359	77
101	0.0017940	0.0023920	0.0023920	0.0029901	0.0029901	0.0029901	0.0059801	0.0059801	0.0059801	0.0059801	79
99	0.0018051	0.0024068	0.0024068	0.0030085	0.0030085	0.0030085	0.0060170	0.0060170	0.0060170	0.0060170	81
97	0.0018140	0.0024187	0.0024187	0.0030233	0.0030233	0.0030233	0.0060466	0.0060466	0.0060466	0.0060466	83
95	0.0018207	0.0024275	0.0024275	0.0030344	0.0030344	0.0030344	0.0060689	0.0060689	0.0060689	0.0060689	85
93	0.0018251	0.0024335	0.0024335	0.0030418	0.0030418	0.0030418	0.0060837	0.0060837	0.0060837	0.0060837	87
91	0.0018273	0.0024364	0.0024364	0.0030456	0.0030456	0.0030456	0.0060911	0.0060911	0.0060911	0.0060911	89
Type C	25.5	29	33	37.5	42.5	47.5	55	65	75	85	Type C
Vert. angles*	Horizontal Mid-zone Angles from the Principal Axis										Vert. angles**

Table notes: *Vertical mid-zone angles above the principal axis **Vertical mid-zone angles below the principal axis

Annex B – Photometric Report and Test Results

A typical multipage form, showing a floodlight photometric report and test results, is presented in this Annex.

Floodlight photometric report, page 1:

(Note: The floodlight photometric report will typically include a computer-generated luminaire sketch similar to the example sketch given on this page. The dimensioned lengths in the example sketch are given in inches.)

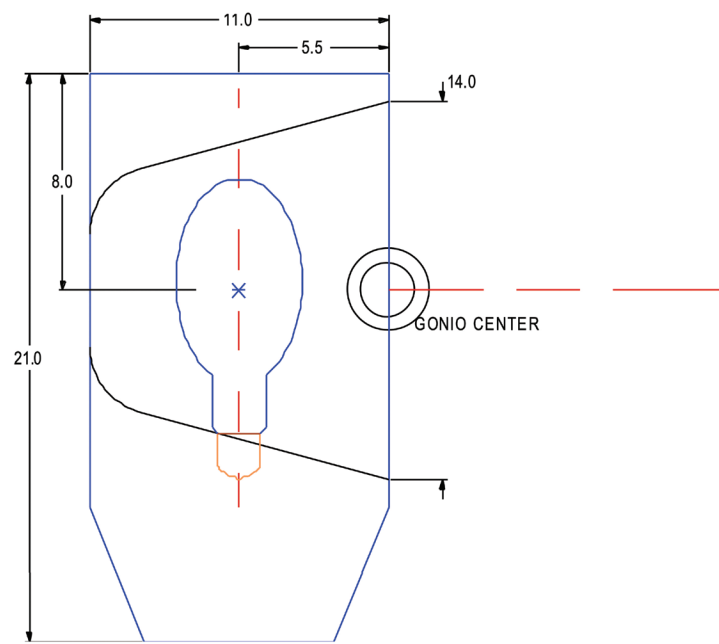
```
+-----+
|   IES DATA REPORT - PHOTOMETRIC LABORATORY   |
+-----+
|           Address    -    Phone                |
+-----+

PHOTOMETRIC DATA for:  TPCEX01                      [ Date ]
```

```
IES TPC EXAMPLE REPORT - TYPE 7H x 7V FLOODLIGHT
FORMED ALUMINUM REFLECTOR; CLEAR FLAT COVER GLASS
400 watt METAL HALIDE LAMP; 36,000 LUMENS; CLEAR ED-37 BULB
SOURCE = 2.0" x .76" dia.;  LCL = 7.0";  MOGUL BASE
```

FLOODLIGHT CHARACTERISTICS

IESNA TYPE	7H x 7V		
MAXIMUM CANDELA	10000.	at	0.0 deg H; 0.0 deg V
BEAM SPREAD:	90. deg Horiz	BEAM LUMENS:	12828.
(at 50% Max CD)	90. deg Vert	Efficiency	35.6 %
FIELD SPREAD:	153. deg Horiz	FIELD LUMENS:	21773.
(at 10% Max CD)	153. deg Vert	Efficiency	60.5 %
FLOODLIGHT WATTS	400.	SPILL LIGHT LUMENS:	205.
LAMP LUMENS	36000.	TOTAL FLOODLIGHT LUMENS:	21978.
		Efficiency	61.0 %



This test was performed in accordance with ANSI/IES LM-35-20 with a test distance of 8 meters (26 feet)

ANGULAR DATA IS RELATIVE TO A HORIZONTAL POLAR AXIS.

Approved Method: Photometric Testing of Floodlights Using High Intensity
Discharge or Incandescent Filament Lamps

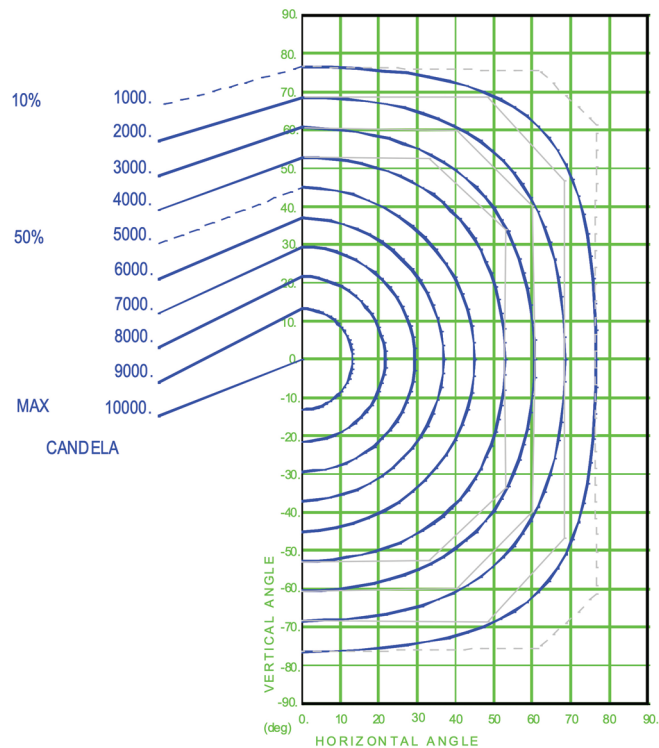
Floodlight photometric report, page 2:

(Note: The photometric report will typically include computer-generated iso-candela curves for the floodlight under test, similar to the example curves given on this page.)

```
+-----+
|   IES DATA REPORT - PHOTOMETRIC LABORATORY   |
+-----+
|           Address      -   Phone              |
+-----+
```

PHOTOMETRIC DATA for: TPCEX01

IES TPC EXAMPLE REPORT - TYPE 7H x 7V FLOODLIGHT
FORMED ALUMINUM REFLECTOR; CLEAR FLAT COVER GLASS
400 watt METAL HALIDE LAMP; 36,000 LUMENS; CLEAR ED-37 BULB
SOURCE = 2.0" x .76" dia.; LCL = 7.0"; MOGUL BASE



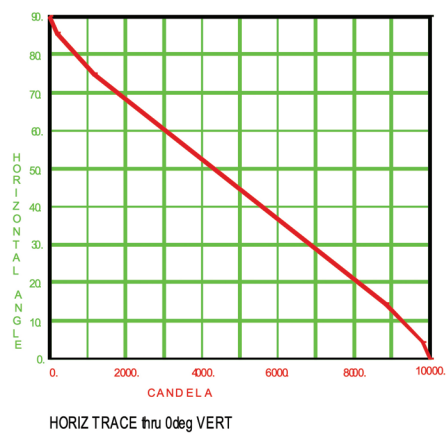
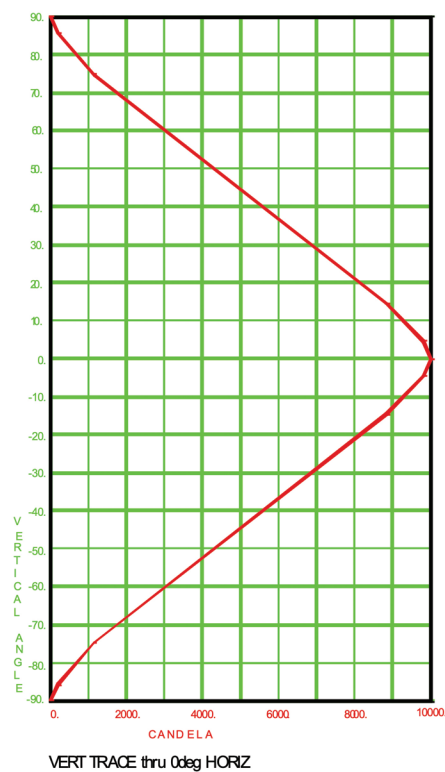
Floodlight photometric report, page 3:

(Note: The photometric report will typically include computer-generated vertical and horizontal axial candela plots for the floodlight under test, similar to the example plots given on this page.)

IES DATA REPORT - PHOTOMETRIC LABORATORY	
Address	Phone

PHOTOMETRIC DATA for: TPCEX01

IES TPC EXAMPLE REPORT - TYPE 7H x 7V FLOODLIGHT
FORMED ALUMINUM REFLECTOR; CLEAR FLAT COVER GLASS
400 watt METAL HALIDE LAMP; 36,000 LUMENS; CLEAR ED-37 BULB
SOURCE = 2.0" x .76" dia.; LCL = 7.0"; MOGUL BASE



ANNEX C – Data File Formatted for Electronic Media

An IES-format data file in an electronic medium is usually supplied for floodlights tested per this Approved Method. *ANSI/IES LM-63-19, IES Standard File Format for Electronic Transfer of Photometric Data* should be followed.⁸

An example of an IES-format data file for a floodlight:

```
IES:LM-63-2019
[TEST] TPCEX01
[TESTLAB] IES TPC
[ISSUEDATE] 9-10-2020
[OTHER] IES TPC EXAMPLE REPORT - TYPE 7H x 7V FLOODLIGHT
[LUMINAIRE] FORMED ALUMINUM REFLECTOR; CLEAR FLAT COVER GLASS
[LAMP] 400 WATT METAL HALIDE LAMP; 36,000 LUMENS; CLEAR ED-37 BULB
[MORE] SOURCE = 2.0" x .76" dia.; LCL = 7.0"; MOGUL BASE
TILT=NONE

1 36000. 1.000 45 23 2 1 1.10 2.10 0.00
1.000 1.000 400.
-90 -85 -75 -65 -55 -47.5 -42.5 -37.5 -33 -29 -25.5 -22.5 -19.5 -17 -15 -13 -11 -9 -7
-5 -3 -1 0
1 3 5 7 9 11 13 15 17 19.5 22.5 25.5 29 33 37.5 42.5 47.5 55 65 75 85 90
0 1 3 5 7 9 11 13 15 17 19.5 22.5 25.5 29 33 37.5 42.5 47.5 55 65 75 85 90
0 250 1143 2429 3714 4679 5322 5965 6544 7058 7508 7894 8280 8601 8858 9114 9371 9550
9650 9750 9850 9950 10000
9950 9850 9750 9650 9550 9371 9114 8858 8601 8280 7894 7508 7058 6544 5965 5322 4679
3714 2429 1143 250 0
0 250 1143 2428 3713 4678 5321 5964 6542 7056 7506 7891 8277 8597 8854 9109 9365 9545
9646 9745 9842 9929 9950
9929 9842 9745 9646 9545 9365 9109 8854 8597 8277 7891 7506 7056 6542 5964 5321 4678
3713 2428 1143 250 0
0 250 1141 2424 3707 4670 5311 5952 6528 7040 7487 7870 8252 8568 8821 9071 9320 9507
9619 9709 9788 9842 9850
9842 9788 9709 9619 9507 9320 9071 8821 8568 8252 7870 7487 7040 6528 5952 5311 4670
3707 2424 1141 250 0
```

ANSI/IES LM-35-20(R2023)

0 249 1138 2416 3694 4653 5291 5929 6501 7008 7450 7827 8202 8511 8756 8997 9233 9435
9570 9647 9709 9745 9750
9745 9709 9647 9570 9435 9233 8997 8756 8511 8202 7827 7450 7008 6501 5929 5291 4653
3694 2416 1138 249 0
0 248 1133 2403 3676 4629 5262 5894 6460 6960 7395 7764 8129 8428 8663 8892 9112 9322
9471 9570 9619 9646 9650
9646 9619 9570 9471 9322 9112 8892 8663 8428 8129 7764 7395 6960 6460 5894 5262 4629
3676 2403 1133 248 0
0 247 1126 2387 3651 4596 5224 5848 6406 6898 7323 7682 8035 8322 8545 8759 8963 9152
9322 9435 9507 9545 9550
9545 9507 9435 9322 9152 8963 8759 8545 8322 8035 7682 7323 6898 6406 5848 5224 4596
3651 2387 1126 247 0
0 245 1118 2366 3620 4556 5176 5791 6340 6821 7235 7583 7922 8195 8405 8605 8793 8963
9112 9233 9320 9365 9371
9365 9320 9233 9112 8963 8793 8605 8405 8195 7922 7583 7235 6821 6340 5791 5176 4556
3620 2366 1118 245 0
0 244 1108 2341 3583 4508 5119 5724 6261 6731 7132 7467 7792 8051 8248 8433 8605 8759
8892 8997 9071 9109 9114
9109 9071 8997 8892 8759 8605 8433 8248 8051 7792 7467 7132 6731 6261 5724 5119 4508
3583 2341 1108 244 0
0 241 1096 2312 3540 4452 5053 5647 6172 6628 7016 7338 7647 7891 8075 8248 8405 8545
8663 8756 8821 8854 8858
8854 8821 8756 8663 8545 8405 8248 8075 7891 7647 7338 7016 6628 6172 5647 5053 4452
3540 2312 1096 241 0
0 239 1083 2280 3491 4389 4979 5560 6071 6513 6887 7195 7489 7719 7891 8051 8195 8322
8428 8511 8568 8597 8601
8597 8568 8511 8428 8322 8195 8051 7891 7719 7489 7195 6887 6513 6071 5560 4979 4389
3491 2280 1083 239 0
0 236 1065 2233 3422 4300 4875 5438 5931 6355 6711 7002 7276 7489 7647 7792 7922 8035
8129 8202 8252 8277 8280
8277 8252 8202 8129 8035 7922 7792 7647 7489 7276 7002 6711 6355 5931 5438 4875 4300
3422 2233 1065 236 0
0 231 1039 2170 3328 4180 4735 5275 5745 6146 6480 6750 7002 7195 7338 7467 7583 7682
7764 7827 7870 7891 7894
7891 7870 7827 7764 7682 7583 7467 7338 7195 7002 6750 6480 6146 5745 5275 4735 4180
3328 2170 1039 231 0
0 226 1010 2098 3223 4045 4578 5094 5540 5918 6229 6480 6711 6887 7016 7132 7235 7323
7395 7450 7487 7506 7508
7506 7487 7450 7395 7323 7235 7132 7016 6887 6711 6480 6229 5918 5540 5094 4578 4045
3223 2098 1010 226 0

**Approved Method: Photometric Testing of Floodlights Using High Intensity
Discharge or Incandescent Filament Lamps**

0 219 972 2004 3086 3871 4377 4864 5281 5631 5918 6146 6355 6513 6628 6731 6821 6898
6960 7008 7040 7056 7058
7056 7040 7008 6960 6898 6821 6731 6628 6513 6355 6146 5918 5631 5281 4864 4377 3871
3086 2004 972 219 0
0 210 923 1884 2911 3651 4125 4577 4962 5281 5540 5745 5931 6071 6172 6261 6340 6406
6460 6501 6528 6542 6544
6542 6528 6501 6460 6406 6340 6261 6172 6071 5931 5745 5540 5281 4962 4577 4125 3651
2911 1884 923 210 0
0 198 862 1733 2695 3381 3817 4230 4577 4864 5094 5275 5438 5560 5647 5724 5791 5848
5894 5929 5952 5964 5965
5964 5952 5929 5894 5848 5791 5724 5647 5560 5438 5275 5094 4864 4577 4230 3817 3381
2695 1733 862 198 0
0 184 786 1549 2431 3055 3448 3817 4125 4377 4578 4735 4875 4979 5053 5119 5176 5224
5262 5291 5311 5321 5322
5321 5311 5291 5262 5224 5176 5119 5053 4979 4875 4735 4578 4377 4125 3817 3448 3055
2431 1549 786 184 0
0 169 703 1347 2146 2706 3055 3381 3651 3871 4045 4180 4300 4389 4452 4508 4556 4596
4629 4653 4670 4678 4679
4678 4670 4653 4629 4596 4556 4508 4452 4389 4300 4180 4045 3871 3651 3381 3055 2706
2146 1347 703 169 0
0 143 566 1056 1684 2146 2431 2695 2911 3086 3223 3328 3422 3491 3540 3583 3620 3651
3676 3694 3707 3713 3714
3713 3707 3694 3676 3651 3620 3583 3540 3491 3422 3328 3223 3086 2911 2695 2431 2146
1684 1056 566 143 0
0 106 364 722 1056 1347 1549 1733 1884 2004 2098 2170 2233 2280 2312 2341 2366 2387
2403 2416 2424 2428 2429
2428 2424 2416 2403 2387 2366 2341 2312 2280 2233 2170 2098 2004 1884 1733 1549 1347
1056 722 364 106 0
0 65 192 364 566 703 786 862 923 972 1010 1039 1065 1083 1096 1108 1118 1126 1133
1138 1141 1143 1143
1143 1141 1138 1133 1126 1118 1108 1096 1083 1065 1039 1010 972 923 862 786 703 566
364 192 65 0
0 22 65 106 143 169 184 198 210 219 226 231 236 239 241 244 245 247 248 249 250 250 250
250 250 249 248 247 245 244 241 239 236 231 226 219 210 198 184 169 143 106 65 22 0
0
0 0 0 0 0 0

INFORMATIVE REFERENCES

- 1 Illuminating Engineering Society. ANSI/IES LM-11-20/R23, Approved Method: Photometric Testing of Searchlights Using Incandescent or HID Sources. New York: IES; 2020.
- 2 Illuminating Engineering Society. ANSI/IES LM-20-20, Approved Method: Photometry of Reflector Type Lamps. New York: IES; 2020.
- 3 Illuminating Engineering Society. ANSI/IES LM-28-20, Approved Method: Guide for the Selection, Care, and Use of Electrical Instruments in the Photometric Laboratory. New York: IES; 2020.
- 4 International Commission on Illumination (CIE). CIE S 017/E:2011, ILV: International Lighting Vocabulary. Vienna: CIE; 2011.
- 5 International Commission on Illumination (CIE). CIE 043-1979, Photometry of Floodlights. Vienna: CIE; 1979.
- 6 Illuminating Engineering Society. ANSI/IES LM-75-19, Approved Method: Guide to Goniometer Measurements and Types, and Photometric Coordinate Systems. New York: IES; 2019.
- 7 McCullough JH, McCullough H. Floodlight photometry without special photometer and without tipping luminaire. Illuminating Engineering. 1967;62(4):243.
- 8 Illuminating Engineering Society. ANSI/IES LM-63-19, IES Standard File Format for Electronic Transfer of Photometric Data. New York: IES; 2019.



Lighting Science Standards

Fundamentals, Metrics and Calculations

Lighting Practice Standards

Design, Engineering, and Specifications

Lighting Applications Standards

Design Criteria and Illumination Recommendations

Lighting Measurements and Testing Procedure Standards

Industry Standardization

Roadway and Parking Facility Lighting Standards

Criteria and Illumination Recommendations

Order# ANSI/IES LM-35-20(R2023)

ISBN# 978-0-87995-350-8

www.ies.org